

Solution Requirements

Date	27 June 2026
Team ID	SWTID-2026-3658
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	The system shall provide secure user registration and login using JWT authentication and encrypted passwords (bcrypt).	High
2	Authorization Levels	The system shall support authenticated users with access to personalized learning features. Future versions may include Admin and Student roles.	Medium
3	External Interfaces	The system integrates with Google Gemini 1.5 Flash API for Question Answering, Quiz Generation, Summarization, and Learning Path Recommendation, and LaMini-Flan-T5 for Concept Explanation.	High
4	Transaction Processing	The system shall accept user queries, process AI requests, generate educational responses, and store user history and bookmarks in the database.	High
5	Reporting	The system shall maintain learning history, quiz results, summaries, and bookmarked responses for future reference.	Medium

6	Business Rules	Users must provide valid educational input. The system validates requests before generating AI responses and stores only authenticated user activities.	High
7	Compliance to Laws or Regulations	The application follows data privacy practices by securely storing user information, encrypting passwords, and protecting API credentials using environment variables.	High
8	Other	The system supports Question Answering, Concept Explanation, Quiz Generation, Text Summarization, Personalized Learning Paths, and response bookmarking.	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should generate networking suggestions quickly and provide fast API responses.	Average response time < 3 seconds
2	Scalability	The system should support additional users and future cloud deployment.	Supports at least 100+ concurrent users in future deployment
3	Security & Data Privacy	User data, feedback, and API interactions should be securely handled.	No sensitive user data stored; secure API communication
4	Reliability & Availability	The application should operate continuously without major failures.	System availability > 95% during testing
5	Usability & Accessibility	The interface should be intuitive, user-friendly, and easy to navigate.	Users can generate suggestions within 3 interaction steps
6	Maintainability & Portability	The code should be modular, reusable, and easy to deploy across different environments.	Modular architecture with reusable services and documented APIs